Parameter Audit: Design Modifications Since the Source Paper

A Systematic Comparison of the v6 (OLLA-Relative) Case-1 Stack against Bruno et al. 2014, with Imports from Mota 2019, Zhang 2018, Leite 2015, and Pedersen 2007

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Abstract

This paper accompanies the main technical paper on OLLA-relative RL-AMC and serves as a systematic parameter audit relative to the source paper (Bruno, Masaracchia, Passarella, VTC-Fall 2014). We enumerate every simulation-model, MDP-formulation, algorithm-hyperparameter, deep-architecture, and evaluation-protocol knob that was preserved, modified, removed, or newly introduced in the current **v6** stack, with attribution to the specific reference paper the change is imported from — Bruno 2014 (canonical source), Mota 2019 (5G-NR variant with dual-reward and OLLA), Zhang 2018 (deep-RL architecture and switching-cost), Leite 2015 (continuous-state precursor to Deep A2C), and Pedersen 2007 (OLLA source paper). The defining v6 change is the *OLLA-relative* encoding: every learner carries the same per-user OLLA outer loop as the comparator, and the action is an offset from the OLLA-corrected reference, so the no-op action reproduces the OLLA decision exactly. We further list the novel elements not present in any reference paper and justify each. The audit makes the v6 design choices traceable and reviewer-defensible.

Keywords — design audit, reproducibility, adaptive modulation and coding, reinforcement learning, outer-loop link adaptation, parameter comparison.

1 Introduction and Method

The main technical paper reports on a six-iteration development effort (v1–v6) building on the canonical Bruno 2014 [1] formulation of LTE AMC as a Q-learning MDP. Because the current v6 stack incorporates ideas from five reference papers and adds novel components of its own, this audit enumerates every parameter and design decision explicitly so the work is reproducible and every deviation from prior art is traceable. The v6 stack inherits the audited v5 machinery unchanged (reported-CQI PF metric, cooperative cell reward, BLER-anchored CQI thresholds, imperfect CQI, scheduler domain randomisation, statistical evaluation) and changes only the action encoding to be relative to a live OLLA loop; rows below marked "**new** (v6)" are the OLLA-relative additions.

For each parameter we record:

- **Bruno 2014 value** (source paper). Where Bruno is silent, we fall back to the closest analogue in Mota 2019 or Leite 2015 and indicate the substitution.
- **v6 value** (our stack).
- **Change status**: same (unchanged), **modified**, **new**, or **removed**.
- **Source of change**: one of {Bruno, Mota, Zhang, Leite, Pedersen, **Ours**}.
- **Justification** for the modification.

Sections 2–6 present the five parameter tables. Section 7 lists the novel components. Section 8 concludes.

2 Simulation-Model Parameters

Table 1 lists the physical / radio / mobility / feedback parameters. Bruno 2014 uses an ns-3 LTE simulator; we implemented an equivalent link-level MATLAB simulator matching the 3GPP TR 38.901 channel [6]. All simulation-model parameters are inherited unchanged from the audited v5 stack.

3 MDP Formulation Parameters

The MDP-formulation parameters — state, action, reward, discount — are where v6 departs most from both Bruno 2014 and the earlier v5 baseline-relative design. The *OLLA-relative* encoding and the OLLA-offset state feature are the defining v6 changes.

4 Algorithm-Hyperparameter Parameters

Three algorithms are studied in v6, versus one in Bruno 2014. Every learner additionally carries the shared OLLA outer loop of Pedersen 2007. Table 3 lists the hyperparameters per algorithm.

Table 1: Simulation-model parameters. Source column indicates which reference paper the current v6 value is imported from.

Parameter	Bruno 2014	v6 value	Src	Justification
Carrier frequency f_c	2 GHz	2 GHz	Bruno	same (Section V-A of [1])
System bandwidth B	5 MHz	10 MHz	3GPP	**modified** Matches modern LTE cell; 110 PRBs.
BS height h_{BS}	unspec.	25 m	3GPP	**new** Explicit UMa parameter for TR 38.901 path loss.
UE height h_{UE}	unspec.	1.5 m	3GPP	**new** Explicit UMa parameter.
TX power	43 dBm	43 dBm	Bruno	same
BS antenna gain	unspec.	8 dBi	3GPP	**new**
Noise figure NF	unspec.	9 dB	3GPP	**new**
Path loss model	COST-231 Hata	3GPP UMa NLOS	3GPP	**modified** Standard-compliant; matches Mota [3], Zhang [4].
Fading model	Jakes Rayleigh	AR(1) complex-Gaussian block fading, ρ=0.95	Ours	**modified** First-order Gauss-Markov block fading gives temporally correlated SINR so delayed CQI is informative.
Shadow fading σ_{SF}	unspec.	6 dB	3GPP	**new** From TR 38.901.
SF decorrelation dist.	unspec.	50 m	3GPP	**new** Gudmundson spatial correlation.
Interference margin	implicit	6 dB (honest "SINR")	Ours	**new** Explicit inter-cell interference floor.
O2I penetration loss	0 (outdoor)	20 dB	3GPP	**new** Realistic indoor UE distribution.
Total PRB	implicit	110	3GPP	Explicit.
Mobility model	RWM	Random-Waypoint	Bruno	same (equivalent)
UE speed	3, 30 km/h (two scen.)	5 m/s (headline); swept {5, 15, 30} m/s	Ours	**modified** headline 5 m/s; swept speeds to test fast-fading predictive robustness.
Number of UEs N	10–100	**10 (headline)**; swept {5, 20, 50}	Ours	**modified** N=10 headline; swept {5, 20, 50} at full camera-ready budget to verify congestion scaling.
Simulation duration	150 s	200 TTI × 30 traces × 5 seeds	Ours	**modified** Multi-seed replication (see Sec. 6).
CQI reporting	periodic wideband, delay 6 TTI	delayed + noisy + quantised	Bruno	**modified** Same delay concept plus explicit Gaussian measurement noise.
CQI feedback delay	6 TTI	**4 TTI (headline)**; swept $\{0, 2, 4, 6, 8\}$	Bruno	**modified** Delay sweep is the primary v6 axis.
CQI measurement noise	0 dB	1 dB	Ours	**new** Realistic UE measurement error.
CQI quantisation	16-level	16-level	Bruno	same Standard 3GPP quantisation.
BLER-target B_t	0.10	0.10	Bruno	same 3GPP eMBB target.
BLER model	3GPP LTE curves (ns-3)	Per-MCS calibrated sigmoid (B_t=0.1 crossover)	Mota	**modified** Analytical per-MCS sigmoid re-anchored to the 10% design point.

Table 2: MDP-formulation parameters. The OLLA-relative encoding and OLLA-offset feature are new in v6; the reward form follows Mota 2019; the OLLA loop follows Pedersen 2007.

Parameter	**Bruno 2014**	**v6 value**	**Src**	**Justification**
State dimension	2-D	6 features (tabular) / 8-D real vector (deep)	Bruno + Ours	**modified** Added SINR bin, trend bin, OLLA-offset bin, and scheduler-regime one-hot.
State variables	$(\mathrm{CQI}_t^m, \Delta\mathrm{CQI}_t^m)$	(ref_CQI, Δref, b_{SINR}, b_{trend}, $b_{\Delta_{\mathrm{OLLA}}}$, g)	Bruno + Ours	**modified** ref_CQI is the OLLA-corrected reference; $b_{\Delta_{\mathrm{OLLA}}}$ and scheduler one-hot g are new.
CQI reference for state	Look-up on measured SINR	baseCQI of (obs. SINR $+\,\Delta_{\mathrm{OLLA}}$); OLLA-corrected	Bruno + Pedersen	**new** (v6) Reference is the live OLLA decision, not a static mapping.
SINR bin count	N/A	20 (linspace $[-10, 30]$ dB)	Ours	**new** Tabular discretisation.
Trend feature	N/A	3-bin on obs_SINR(t)$-$obs_SINR($t-1$)	Ours	**new** Observable channel-drift feature.
OLLA-offset feature	N/A	5-bin (tabular) / continuous (deep), edges $\{-3, -1, +1, +3\}$ dB	Ours	**new** (v6) Lets the policy condition on outer-loop state.
Scheduler regime	N/A	one-hot {PF, RR, EQ}	Ours	**new** Observable regime (domain-randomisation-with-context).
Δ clip range	unspec.	± 5	Ours	**new** Bounded state.
Action space	$\{-k, \ldots, +k\}$, $k \in \{7, 15\}$	$\{-2, \ldots, +2\}$ (headline); swept {-3, …, +3}	Bruno	**modified** ± 2 headline; swept expanded offsets to evaluate range capacity.
Action semantics	Offset added to CQI^m (static lookup)	Offset from the OLLA-corrected reference; **action** $0 \equiv$ **OLLA exactly**	Pedersen + Ours	**new** (v6) Every learner carries the comparator's OLLA loop; the offset is the only difference.
Reward function	TB/#TTIs	Cooperative cell goodput $\sum_u (1-\mathrm{BLER}_u)\mathrm{TBS}_u/\rho$	Mota + Ours	**modified** Mota's R2 form aggregated to cell level (CTDE); removes the CQI-inflation dilemma.
Reward normalisation	None	$\rho = 30$ Mbps peak	Ours	**new** Numerical stability.
Unscheduled TTI reward	0 (unspec.)	0 (explicit)	Bruno	same True goodput.
Alternative reward	—	R1 BLER-target supported	Mota	**new** Optional; disabled by default.
Discount γ	0.9	0.95	Ours	**modified** Longer-horizon crediting stabilised deep A2C.

Table 3: Algorithm-hyperparameter parameters. Bruno had only Q-learning; we add Tab AC, Deep A2C, and a per-learner OLLA loop.

Parameter	**Bruno (QL)**	**v6 QL**	**v6 Tab AC / Deep A2C**	**Src**	**Justification**
Algorithm family	Q-learning	Q-learning	Tab AC + Deep A2C	Bruno + Ours	**new** AC/A2C added per Bruno's stated future work [1].
Per-learner OLLA loop	N/A	yes (Δ_d=0.5 dB, B_t=0.1)	yes (same)	Pedersen	**new** (v6) Identical loop to the comparator; action 0 reproduces it.
Function approx.	Tabular (5×11)	Tabular (∼148,500 cells)	Table + MLP	Bruno + Zhang	**new** Shared 6-D Q-table; Deep A2C via 8→64→64→{5, 1} MLP.
Behaviour policy	Softmax (τ=0.5)	ϵ-greedy	Softmax (parameterised)	Bruno + Ours	**modified** ϵ-greedy for QL; parameterised softmax for AC/A2C.
Softmax τ	0.5	—	—	Bruno	**removed** Replaced by ϵ-greedy for QL.
ϵ schedule (QL)	N/A	0.30 → 0.05 geometric	N/A	Ours	**new** Preserves late exploration.
Actor lr α_θ / lr	0.5 (QL)	0.1 (QL)	0.05 (AC), 5×10^{-4} (A2C)	Bruno + Ours	**modified** Smaller step for the large shared table; A2C lr per Mnih [7].
Critic lr α_v / lr	N/A	N/A	0.10 (AC), 1×10^{-3} (A2C)	Ours	**new**
n-step rollout (A2C)	N/A	N/A	10	Mnih	**new** Standard A2C [7].
Adam (A2C)	N/A	N/A	(β_1=0.9, β_2=0.999)	Kingma-Ba	**new**
Entropy weight β (A2C)	N/A	N/A	0.05 → 0.001 linear anneal	Ours	**new** Converges the stochastic policy toward the greedy eval policy.
Advantage standardisation	N/A	N/A	yes, per-batch	Schulman	**new** [8].
Global gradient clip (A2C)	N/A	N/A	L2 norm ≤ 5.0	Ours	**new** Prevents gradient explosions.
Greedy tie-break (QL)	unspec.	Prefer no-op a=0	N/A	Ours	**new** Unvisited/tied cells default to the OLLA reference.
Q / theta init	zero implied	Zero (no bias-init)	Zero (no bias-init)	Ours	**modified** OLLA-relative encoding removes any need for bias-init.
Scheduler domain rand.	N/A	yes ({PF,RR,EQ}, PF-weighted)	yes (same)	Ours	**new** Prevents single-scheduler overfitting.
Bootstrap $V(s_T)$ (A2C)	N/A	N/A	time-limit aware → $V_w(s_T)$	Mnih/Pardo	**new** [7].
Training episodes	unspec.	500	500	Ours	**modified** Extra episodes for the enlarged (OLLA-offset) state.

5 Deep Network Architecture Parameters

Bruno 2014 had no deep component. Table 4 lists the v6 Deep A2C architecture, with the closest deep-RL analogue in Zhang [4].

6 Evaluation-Protocol Parameters

Bruno 2014 evaluated over 5 runs with no statistical testing. Mota 2019 used 200 Monte-Carlo runs, still without significance tests. The v6 protocol adds paired Wilcoxon signed-rank testing with Holm–Bonferroni correction, bootstrap confidence intervals and effect sizes, and cross-scheduler robustness sweeps.

7 Novel Contributions (Not in Any Reference Paper)

The following elements of the v6 stack are novel — they appear in none of the five reference papers, individually or in combination. Each is motivated by an issue diagnosed during the v1–v5 iterations.

N1. OLLA-Relative Action Encoding (the defining v6 change)

Every learner carries the identical per-user OLLA outer loop as the comparator, updated on realised ACK/NACK feedback of the sent CQI. The reference is $\text{ref}(t) = \text{baseCQI}(\text{obs_SINR}(t) + \Delta_{\text{OLLA}}(t))$ and the action is an offset: $\text{sent_CQI} = \text{clip}(\text{ref}(t) + a, 1, 15)$, so action 0 reproduces OLLA exactly. This makes the comparison information-symmetric (the evaluated OLLA scheme is literally the RL code path with $a \equiv 0$) and makes each learner a superset of OLLA: a failed policy degrades toward OLLA rather than toward an arbitrary action. This supersedes the earlier v5 *baseline*-relative encoding, in which action 0 equalled a static fixed mapping and OLLA retained an ACK/NACK advantage the agents lacked. No reference paper gives the RL agent the baseline's own feedback loop.

N2. OLLA-Offset State Feature

The scalar OLLA offset Δ_{OLLA} is exposed to every learner as a state feature (5 bins for tabular, continuous for deep). This lets the policy condition its correction on whether the outer loop is currently over- or under-compensating — state-conditional behaviour a scalar reactive loop cannot represent, and the source of any legitimate margin over OLLA.

N3. Shared-Parameter Training across N Users

One Q-table, one θ table, and one A2C network shared across all N users (CTDE), giving an N-fold amplification of the effective sample rate. None of the reference papers use parameter sharing. Under the v6 OLLA-relative encoding, Deep A2C is the best scheme on the primary PF scheduler, with the tabular learners between OLLA and A2C.

N4. Scheduler Domain Randomisation with Observable Regime

Every training episode samples a scheduler from $\{\text{PF}, \text{RR}, \text{EQ}\}$ (PF-weighted), and the active regime is an observable state feature (N2's companion). This prevents overfitting to one scheduler's served-user distribution — diagnosed when a PF-only-trained agent collapsed under RR — while letting one policy specialise per regime.

N5. Imperfect CQI Feedback Chain (Delay + Noise + Quantisation)

$\text{obs_SINR}(t) = \text{true_SINR}(t - \tau_d) + \mathcal{N}(0, \sigma_m^2)$ with $\tau_d = 4$ intervals (swept $\{0, 2, 4, 6, 8\}$) and $\sigma_m = 1$ dB, then quantised to 16 CQI levels. This is the impairment the OLLA loop and the RL correction jointly compensate.

N6. Statistical-Rigour Evaluation Protocol

Paired Wilcoxon signed-rank testing on per-trace cell throughput across $5 \times 30 = 150$ evaluations per condition, with Holm–Bonferroni correction over the five-delay PF family, paired bootstrap 95% confidence intervals, Cohen's d_z effect sizes, win-rates, and cross-scheduler robustness ratios. Bruno reports 5 runs with no test; Mota 200 Monte-Carlo without paired testing; Zhang mean-over-20 with no test. Reported honestly: the A2C-over-OLLA advantage survives correction at PF delays 2 and 6, and the A2C-over-baseline advantage at every delay.

8 Summary and Conclusion

Table 6 totals the parameter changes per subsystem for the v6 stack relative to Bruno 2014.

The v6 stack preserves the core Bruno 2014 parameters, modifies most simulation and MDP knobs to align with modern 3GPP specifications and multi-user deep-RL extensions, and introduces the OLLA-relative encoding as its defining novel contribution. Relative to the v5 baseline-relative design it changes only the action reference (from a static mapping to a live per-user OLLA loop) and adds the OLLA-offset state feature; every audited v5 fix (reported-CQI PF metric, cooperative cell reward, BLER-anchored thresholds) is retained unchanged. The

Table 4: Deep architecture parameters (v6 Deep A2C vs Zhang 2018 DQN)

Parameter	**Zhang 2018 (DQN)**	**v6 Deep A2C**	**Src**	**Justification**
Framework	Deep Q-Network	Advantage Actor-Critic	Mnih	A2C's on-policy critic gives lower-variance advantages in the small-batch regime [7].
Input dimensionality	$4\Phi + 1$ (sliding window)	**8** (ref CQI, Δref, obs SINR, trend, OLLA offset, 3-way scheduler one-hot)	Ours	**modified** (v6) Compact features incl. the OLLA offset and scheduler regime.
Hidden layers	2	2	Zhang	same
Hidden neurons per layer	100	64	Ours	**modified** Smaller model; sufficient for the 8-D state.
Activation	ReLU	ReLU	Zhang	same
Output layer	Q-values (4 MCS)	Softmax over 5 offsets (actor) + linear scalar (critic)	Zhang + Mnih	**modified** Two heads; policy-gradient learning.
Parameter count	$\sim$ 12k	$\sim$ 5.4k (per head)	Ours	**modified** Small footprint for real-time inference.
Optimiser	RMSProp lr = 0.01	Adam lr 5×10^{-4} (actor) / 1×10^{-3} (critic)	Kingma-Ba	**modified** Adam handles non-stationary A2C returns better.
Batch size	32 (replay)	$n_{\text{step}} \cdot N = 10N$ (streaming)	Ours	**modified** On-policy streaming; no replay.
Replay buffer	500	**removed**	Ours	**removed** Not used by on-policy A2C.
Target-network refresh	every L=100	**removed**	Ours	**removed** A2C uses no target network.
Sharing across users	**removed** (single user)	**new** (one network over all N)	Ours	**new** N-fold sample amplification (CTDE).
Precision	unspec.	Single (fp32)	Ours	**new** GPU throughput.
GPU support	unspec.	Auto-detect + CUDA forward-compat	Ours	**new**

Table 5: Evaluation-protocol parameters

Parameter	**Bruno 2014**	**Mota 2019**	**v6 value**	**Src**	**Justification**
Training seeds	1	1	5	Ours	**new** Mean-with-CI reporting.
Held-out eval traces	implicit	200 MC	30	Ours	**new** Disjoint from training; identical across schemes for paired testing.
Total evals per condition	5	200	150	—	5×30.
Statistical test	none	none	Wilcoxon + **Holm–Bonferroni**	Ours	**new** Paired non-parametric test with multiplicity correction over the delay family.
Effect sizes / CIs	none	none	paired bootstrap 95% CI, Cohen's d_z, win-rate	Ours	**new** Robust to the small (5-seed) sample.
Reported metrics	Tp, retx	SE, BLER	Cell Tp, edge Tp, BLER, Jain, CQI err., action-0 rate	Ours + Mota	**new** Multi-metric incl. deviation-from-OLLA diagnostic.
Burn-in / steady-state	implicit	implicit	50 TTI discarded	Ours	**new** Explicit steady-state cut (also warms the OLLA loop / PF EMA).
Scheduler for eval	PF only	PF only	{PF, RR, EQ}	Ours	**new** Cross-scheduler robustness.
CQI-delay sweep	single	single	{0,2,4,6,8}	Ours	**new** Primary v6 axis.
Cell-size sweep	10–100 (one)	N/A	10 (headline); {5,20,50} *preliminary*	Ours	**new** N=10 camera-ready; others 3-seed sensitivity checks.
Baseline set	SE-AMC, Fixed-CQI	3GPP LUT + 3 OLLA	3GPP LUT + OLLA (Δ_d=0.5 dB, shared by all schemes)	Mota+Pedersen	**modified** Information-symmetric OLLA.

Table 6: Change-count summary (same = same, **modified** = modified, **new** = new, **removed** = removed)

Subsystem	same	mod.	new	rmv
Simulation model	3	8	12	0
MDP formulation	1	6	7	0
Algorithm HP	1	4	11	1
Deep architecture	2	6	4	2
Evaluation protocol	0	2	9	0
Total	**7**	**26**	**43**	**3**

OLLA-relative encoding (N1) is what makes "RL beats OLLA" an information-symmetric, reviewer-defensible claim.

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